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Choose values by coordinates.

One tensor. Five rows and five columns.

> Indices start at 0; the floating-point values keep .0.

x: five rows × five columns

	0	1	2	3	4
0	1.0	2.0	3.0	4.0	5.0
1	6.0	7.0	8.0	9.0	10.0
2	11.0	12.0	13.0	14.0	15.0
3	16.0	17.0	18.0	19.0	20.0
4	21.0	22.0	23.0	24.0	25.0

First coordinate: row · second coordinate: column

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A row fixes the first coordinate.

The colon includes every column in that row.

> Fix the row, then include every column.

x[0, :]

Original x

	0	1	2	3	4
0	1.0	2.0	3.0	4.0	5.0
1	6.0	7.0	8.0	9.0	10.0
2	11.0	12.0	13.0	14.0	15.0
3	16.0	17.0	18.0	19.0	20.0
4	21.0	22.0	23.0	24.0	25.0

Selected inputs → outputs

From [row,col]	Input	× 2 result
[0,0]	1.0	?
[0,1]	2.0	?
[0,2]	3.0	?
[0,3]	4.0	?
[0,4]	5.0	?

5 selected inputs · 5 output values

Read each [row, column] coordinate in the original x.

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Multiply the selected row.

Each selected value has one corresponding result.

> Follow each input to its own output.

x[0, :] * 2

Original x

	0	1	2	3	4
0	1.0	2.0	3.0	4.0	5.0
1	6.0	7.0	8.0	9.0	10.0
2	11.0	12.0	13.0	14.0	15.0
3	16.0	17.0	18.0	19.0	20.0
4	21.0	22.0	23.0	24.0	25.0

Selected inputs → outputs

From [row,col]	Input	× 2 result
[0,0]	1.0	2.0
[0,1]	2.0	4.0
[0,2]	3.0	6.0
[0,3]	4.0	8.0
[0,4]	5.0	10.0

5 selected inputs · 5 output values

Read each [row, column] coordinate in the original x.

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A column fixes the second coordinate.

The colon now includes every row.

> Include every row, then fix the column.

x[:, 0]

Original x

	0	1	2	3	4
0	1.0	2.0	3.0	4.0	5.0
1	6.0	7.0	8.0	9.0	10.0
2	11.0	12.0	13.0	14.0	15.0
3	16.0	17.0	18.0	19.0	20.0
4	21.0	22.0	23.0	24.0	25.0

Selected inputs → outputs

From [row,col]	Input	× 2 result
[0,0]	1.0	?
[1,0]	6.0	?
[2,0]	11.0	?
[3,0]	16.0	?
[4,0]	21.0	?

5 selected inputs · 5 output values

Read each [row, column] coordinate in the original x.

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Multiply the selected column.

Same multiplier. Different selected inputs.

> The selection changes; the multiplier remains 2.

x[:, 0] * 2

Original x

	0	1	2	3	4
0	1.0	2.0	3.0	4.0	5.0
1	6.0	7.0	8.0	9.0	10.0
2	11.0	12.0	13.0	14.0	15.0
3	16.0	17.0	18.0	19.0	20.0
4	21.0	22.0	23.0	24.0	25.0

Selected inputs → outputs

From [row,col]	Input	× 2 result
[0,0]	1.0	2.0
[1,0]	6.0	12.0
[2,0]	11.0	22.0
[3,0]	16.0	32.0
[4,0]	21.0	42.0

5 selected inputs · 5 output values

Read each [row, column] coordinate in the original x.

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Change only the index.

Index one means the second row or column.

> Inspect a second row and a second column.



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Pause: interpret this expression.

Predict the selected inputs before multiplying.

> Pause before checking the next slide.

x[:, 3] * 2

Original x

	0	1	2	3	4
0	1.0	2.0	3.0	4.0	5.0
1	6.0	7.0	8.0	9.0	10.0
2	11.0	12.0	13.0	14.0	15.0
3	16.0	17.0	18.0	19.0	20.0
4	21.0	22.0	23.0	24.0	25.0

Selected inputs → outputs

From [row,col]	Input	× 2 result
?	?	?
?	?	?
?	?	?
?	?	?
?	?	?

Which dimension is fixed? Which does the colon include?

Read each [row, column] coordinate in the original x.

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The second coordinate fixes the column.

All rows. Column three. Five selected inputs.

> Return to Chapter 3 and change one selection at a time.

x[:, 3] * 2

Original x

	0	1	2	3	4
0	1.0	2.0	3.0	4.0	5.0
1	6.0	7.0	8.0	9.0	10.0
2	11.0	12.0	13.0	14.0	15.0
3	16.0	17.0	18.0	19.0	20.0
4	21.0	22.0	23.0	24.0	25.0

Selected inputs → outputs

From [row,col]	Input	× 2 result
[0,3]	4.0	8.0
[1,3]	9.0	18.0
[2,3]	14.0	28.0
[3,3]	19.0	38.0
[4,3]	24.0	48.0

5 selected inputs · 5 output values

Read each [row, column] coordinate in the original x.

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